Smart Factory

Monitoring System

IoT-Based Industrial Monitoring & Control

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Internet of Things (IoT)

What is IoT?

Internet of Things — Overview

The **Internet of Things (IoT)** refers to the network of physical objects — "*things*" — embedded with sensors, software and connectivity that enables them to collect and exchange data over the internet **without human-to-human or human-to-computer interaction**.

Things
Physical devices & sensors

Connect
Network & protocols

Collect
Data gathering & storage

Act

IoT — Definition

Key Concepts & Components

"IoT is a system of interrelated computing devices, mechanical and digital machines, objects or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human interaction."

Sensors & Actuators
Perceive the physical world — temperature, motion, light — and trigger actions based on data.

Connectivity
Protocols like MQTT, HTTP, Zigbee, Wi-Fi and Ethernet link devices to the cloud.

Cloud / Edge Platform
Aggregates data, runs analytics, manages devices and triggers automated rules.

User Interface
Dashboards, mobile apps and REST APIs let users monitor and control the system.

IoT Applications & Use Cases

Where IoT is Transforming Industries

Smart Manufacturing

Predictive maintenance, automated quality control, real-time floor monitoring

Smart Homes

Automated lighting, smart thermostats, security systems, voice assistants

Healthcare

Patient monitoring, wearables, smart hospital equipment, drug tracking

Smart Transport

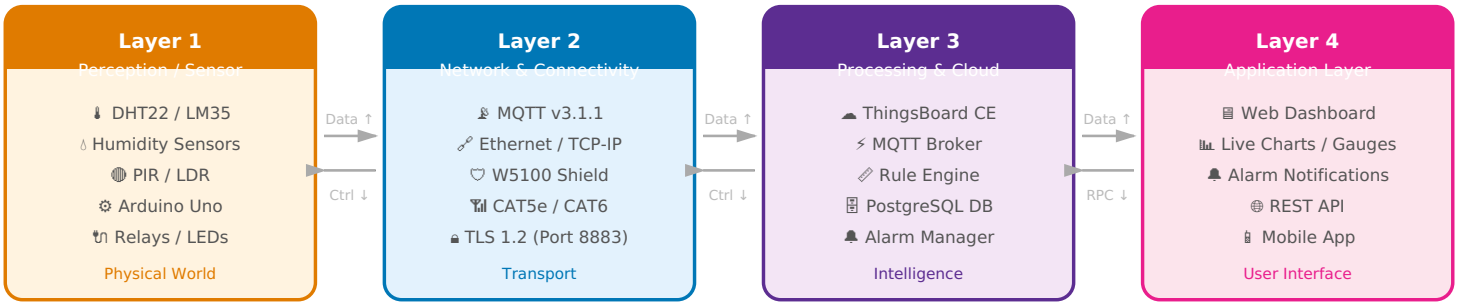
Fleet tracking, autonomous vehicles, traffic management, EV charging

Agriculture

Soil moisture monitoring, precision irrigation, livestock tracking

IoT Architecture

4-Layer Reference Model — Horizontal Flow



Sensor Layer → Network → Cloud Processing → Application | Bidirectional: Data up ↑ Control down ↓

General Use Case — Smart IoT Factory

How IoT Powers Modern Industrial Facilities

Environment Monitoring
Continuous sensing of temperature, humidity and air quality across all factory zones.

Predictive Maintenance
Vibration and current sensors detect anomalies early, preventing machine failures.

Safety & Emergency Response
Sensors automate emergency shutdowns and alerts to protect workers and equipment.

Production Visibility
Real-time OEE dashboards track machine uptime, output rates and bottlenecks.

Remote Control
Engineers remotely toggle relays, conveyors and HVAC systems via web dashboards.

Trend Analysis
Historical data enables energy optimization and regulatory compliance reporting.

Problem Statement

Problem Statement

Industrial Challenges

No Real-Time Visibility

Factory floors lack live data on temperature, humidity and vibration — making early hazard detection impossible.

Delayed Incident Response

Without automated alerting, threshold breaches go unnoticed until physical damage or safety incidents occur.

No Remote Equipment Control

Operators must be physically present to toggle equipment — inefficient and risky during emergencies.

Reactive Maintenance

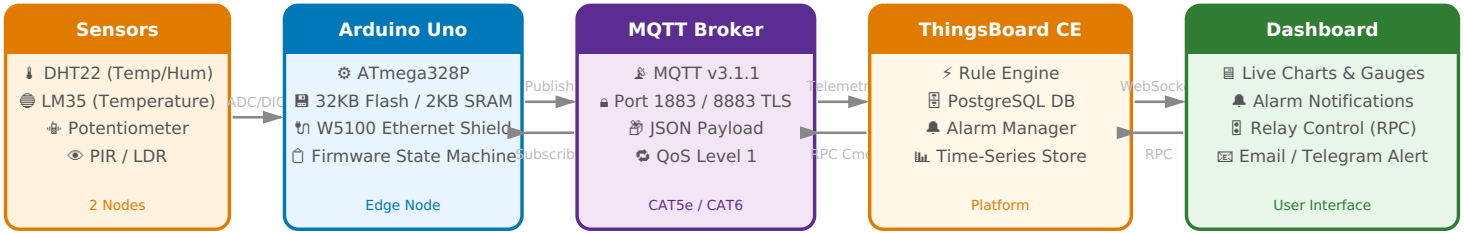
Without trend data, factories rely on breakdown-based maintenance, increasing downtime and costs.

Proposed Solution

Proposed Solution

Distributed IoT Monitoring — Solution Block Diagram

A distributed, multi-node IoT system using **Arduino Uno nodes**, **MQTT over Ethernet** and **ThingsBoard** for real-time monitoring, alerting and remote control.



Sensors → Arduino Uno → MQTT Broker → ThingsBoard → Dashboard | Bidirectional control via RPC

System Architecture

System Architecture

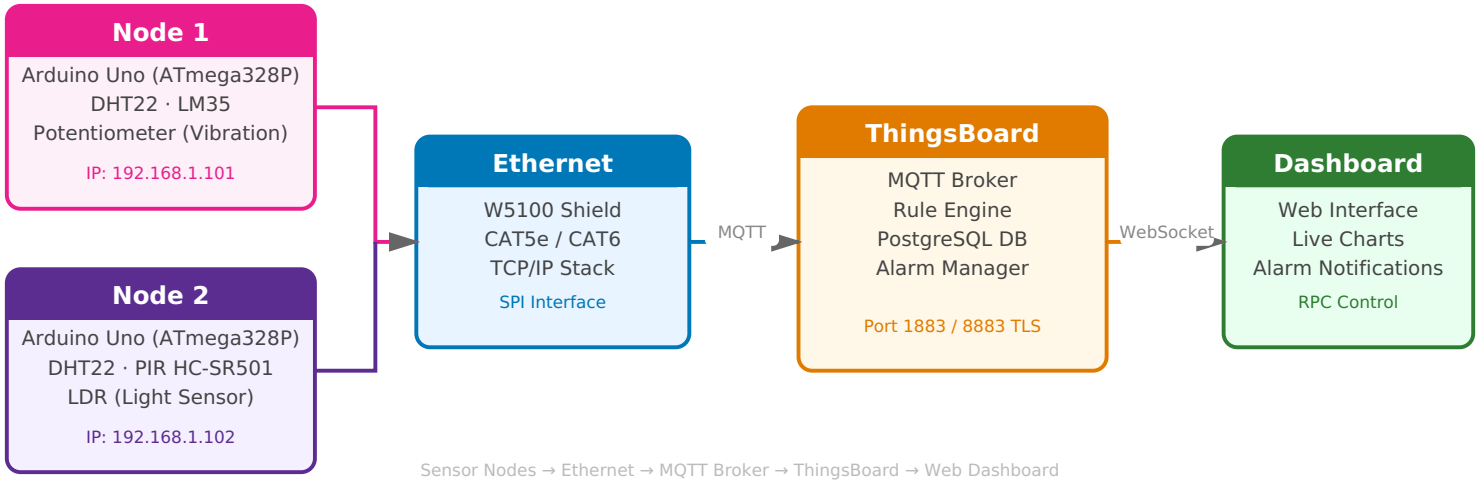
3-Tier IoT Architecture

Presentation Layer	Web Dashboard · REST API · Mobile App
Application Layer	ThingsBoard · Rule Engine · Alarm Manager · Time-Series DB
Edge Layer	Node 1: Production Line · Node 2: Warehouse Monitor
Sensor Layer	DHT22 · LM35 · PIR · LDR

Star topology · Ethernet CAT5e/6 · Static IP Addressing

System Block Diagram

End-to-End Data & Control Flow



Monitoring Nodes

Monitoring Nodes

2-Node Deployment

Node 1

Production Line Monitor

IP: 192.168.1.101

SENSORS

- DHT22 (Temp/Humidity)
- LM35 (Temperature)
- Potentiometer (Vibration)

ACTUATORS

- Relay
- Green / Yellow / Red LEDs
- Push Button

Node 2

Warehouse Monitor

IP: 192.168.1.102

SENSORS

- DHT22 (Temp/Humidity)
- LDR (Light Sensor)
- PIR HC-SR501 (Motion)

Project Requirements

Project Requirements

Functional Requirements

- Real-time sensor data collection every 5 seconds
- Automated threshold-based alerting (temperature, vibration, humidity)
- Remote relay / actuator control via ThingsBoard dashboard
- Historical trend analysis and data reporting
- Multi-node coordination for emergency scenarios
- MQTT telemetry publishing with JSON payload format

Project Requirements

Non-Functional & Simulator Requirements

Non-Functional

- End-to-end latency < 5 seconds
- System uptime > 99%
- Scalable to 100+ nodes
- MQTT message size ≤ 256 bytes
- Dashboard load time < 3 seconds

Simulator (PICSimLab)

- PICSimLab for Arduino Uno simulation
- Virtual sensors: DHT22, LM35, PIR, LDR
- Simulated Ethernet (W5100) connectivity
- No physical hardware required
- Supports full firmware testing & MQTT

Communication Design

Communication Design

MQTT Protocol Implementation

Protocol: MQTT v3.1.1 over TCP/IP (Ethernet)

Broker	**ThingsBoard (1883 / 8883 TLS)**
QoS Level	**1 — At Least Once Delivery**
Keep-Alive	**60 seconds**
Payload Format	**JSON (~150 bytes/msg)**
Publish Rate	**Every 5 seconds/node**
Total Throughput	**60 bytes/sec (2 nodes)**
Reconnect Strategy	**Exp. backoff (5s → 60s)**
Authentication	**256-bit device tokens**

Automated Alert Rules

Automated Alert Rules

Rule Engine Configuration

Temperature > 35°C

CRITICAL

- → Create alarm on dashboard
- → Send email notification
- → Send Telegram alert
- → Red LED on Node

Vibration > 700 (ADC)

WARNING

- → Create vibration alarm
- → Yellow LED indicator
- → Log event to ThingsBoard

Humidity > 80%

WARNING

- → Create humidity alarm
- → Dashboard alert triggered
- → Email notification sent

Thank You

Smart Factory Monitoring System

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